Applied Math and Machine Learning Basics (1)

Gwangsu Kim

JBNU Dep. of Stat.

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Special Lecture for Machine Learning

Outline

Notations I

1. Overview
   - The Matrix Cookbook (Petersen and Pedersen, 2006).
   - Basically, we manipulate the scalars, vectors, and matrices. Furthermore, tensors will be considered.

Notations II

2. Scalars, vectors, matrices, and tensors

- **Scalar**:
  1. Single number (real number or natural number).
  2. Example: Let $s \in \mathbb{R}$ be the slope of the line while defining a real-valued scalar, or Let $n \in \mathbb{N}$ be the number of units while defining a natural number scalar.
  3. In the above, the roles of scalars are defined.
- **Vector**:
  1. A (1-D) array of numbers arranged in order.
  2. A vector $\boldsymbol{x}$ lies in the set formed by taking the Cartesian product of $\mathbb{R}$ n times, denoted as $\mathbb{R}^n$.

Notations III

$$\boldsymbol{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$$

3. x_k denotes the k-the elements, and $\boldsymbol{x}_{\mathcal{S}}$ denotes the vector having elements of $\mathcal{S} = \{s_1, s_2, \ldots, s_l\}$.
   index 집합
4. $\boldsymbol{x}_{-\mathcal{S}}$ denotes the vector having elements of $\mathcal{S}^c$.
5. For example, when $\mathcal{S} = \{3, 4\}$ and $n = 10$, $x_{\mathcal{S}} = (x_3, x_4)^\intercal$ and $x_{-\mathcal{S}} = (x_1, x_2, x_5, x_6, x_7, x_8, x_9, x_{10})^\intercal$.

Notations IV

- **Matrix**:
  1. 2D-array numbers, interpreted as the stack of vectors.
  2. The $\boldsymbol{A} \in \mathbb{R}^{m \times n}$ where $\boldsymbol{A}_{:,k}$ is the vector of the k-th column of $\boldsymbol{A}$.
  3. For example,

$$\boldsymbol{A} = \begin{bmatrix} A_{1,1} & A_{1,2} \\ A_{2,1} & A_{2,2} \end{bmatrix}.$$

*열단위 vector stack
이 중요

  4. $f(\boldsymbol{A})_{i,j}$ is the (i,j) element after the function f for $\boldsymbol{A}$.

$f(x) = y$ 함수
$(f) = y$ functional
$(f) = g$ operator

- **Tensor**:
  1. Array more than two axes.
  2. **A** denote a tensor and $\mathrm{A}_{i,j,k}$ denotes the element of the tensor.

Transpose & Multiplication I

1. Transpose

$$\boldsymbol{A}^{\top}_{i,j} = \boldsymbol{A}^{\top}_{j,i}$$

$$\boldsymbol{A} = \begin{bmatrix} A_{1,1} & A_{1,2} \\ A_{2,1} & A_{2,2} \\ A_{3,1} & A_{3,2} \end{bmatrix} \Rightarrow \boldsymbol{A}^{\top} = \begin{bmatrix} A_{1,1} & A_{2,1} & A_{3,1} \\ A_{1,2} & A_{2,2} & A_{3,2} \end{bmatrix}$$

2. Addition

- $\boldsymbol{A} + \boldsymbol{B} = \boldsymbol{C}$ where $C_{i,j} = A_{i,j} + B_{i,j}$, $a \cdot \boldsymbol{A} + c = \boldsymbol{G}$ where $G_{i,j} = a \cdot A_{i,j} + c$.
- $\boldsymbol{A} + \boldsymbol{b} = \boldsymbol{H}$ where $H_{i,j} = A_{i,j} + b_j$
- Broad casting of $\boldsymbol{b}$: $\boldsymbol{B}$ where $B_{ij} = b_j$.

Transpose & Multiplication II

3. Multiplication

- Multiplication of matrices $\boldsymbol{A} \in \mathbb{R}^{a_1 \times b}$ and $\boldsymbol{B} \in \mathbb{R}^{b \times a_2}$

$$\boldsymbol{C} = \boldsymbol{A}\boldsymbol{B}$$

where $C_{i,j} = \sum_l A_{i,l} B_{l,j}, \ (i,j) = \{1, \dots, a_1\} \times \{1, \dots, a_2\}$.

- * Hadamard product (element-wise product) of $A \in \mathbb{R}^{\underline{a \times b}}$ and $B^{\underline{a \times b}}$

$$\boldsymbol{C} = \boldsymbol{A} \odot \boldsymbol{B}$$

$$\begin{array}{cc} \times w_1 & \times w_2 \\ \multicolumn{2}{c}{\begin{pmatrix} 3 & 6 \\ 9 & 7 \end{pmatrix}} \\ \times w_7 & \times w_3 \end{array}$$

where $C_{i,j} = A_{i,j} B_{i,j}$.

→ DNN에 값 그대로 사용하고 싶지 않을때
(eg: 가중치 조절 등등)
masking

Transpose & Multiplication III

4. Properties of multiplication
   - $\boldsymbol{A}(\boldsymbol{B}+\boldsymbol{C}) = \boldsymbol{A}\boldsymbol{B}+\boldsymbol{A}\boldsymbol{C}, \;\; \boldsymbol{A}(\boldsymbol{B}\boldsymbol{C}) = (\boldsymbol{A}\boldsymbol{B})\boldsymbol{C}.$
   - $\boldsymbol{A}\boldsymbol{B} \neq \boldsymbol{B}\boldsymbol{A}$ (not cummutative), $\;\boldsymbol{x}^\intercal\boldsymbol{y} = \boldsymbol{y}^\intercal\boldsymbol{x}$ and $(\boldsymbol{A}\boldsymbol{B})^\intercal = \boldsymbol{B}^\intercal\boldsymbol{A}^\intercal$.

5. Linear equation
   - $\boldsymbol{A}\boldsymbol{x} = \boldsymbol{b}$ where $\boldsymbol{A} \in \mathbb{R}^{m\times n}, \boldsymbol{b} \in \mathbb{R}^{m\times 1}$.

$$\boldsymbol{A}_{1,:}\boldsymbol{x} = b_1, \;\; \boldsymbol{A}_{2,:}\boldsymbol{x} = b_2 \;\ldots, \;\; \boldsymbol{A}_{m,:}\boldsymbol{x} = b_m.$$

Identity and Inverse Matrix I

- Identity matrix: all diagonal terms are one, others are zero such that $\boldsymbol{I}_n \in \mathbb{R}^{n\times n}$.

$$\forall \boldsymbol{x} \in \mathbb{R}^n,\ \boldsymbol{I}_n \boldsymbol{x} = \boldsymbol{x}.$$

- Inverse matrix of $\boldsymbol{A} \in \mathbb{R}^{m\times n}$ (we can let $m = n$)

$$\boldsymbol{A}^{-1}\boldsymbol{A} = \boldsymbol{I}_n$$

Application in the linear equation ($\boldsymbol{A} \in \mathbb{R}^{m\times n}$ and $\boldsymbol{b} \in \mathbb{R}^m$).

$$\begin{aligned} \boldsymbol{A}\boldsymbol{x} &= \boldsymbol{b} \\ \boldsymbol{x} &= \boldsymbol{A}^{-1}\boldsymbol{b} \end{aligned} \tag{1}$$

* Usually, $\boldsymbol{A}^{-1}$ can be obtained with a limited precision due to the computation problem.

Identity and Inverse matrix I

1. Linear equation
   - The existence of $\boldsymbol{A}^{-1}$ determine the uniqueness of the solution of Eqn. 1.
   - Can have infinitely many or no solutions.
   - For example, $\boldsymbol{x}$ and $\boldsymbol{y}$ are solutions, then $\forall \alpha \in [0, 1], \boldsymbol{z} = \alpha \boldsymbol{x} + (1 - \alpha)\boldsymbol{y}$ are solutions.

2. Linear dependency
   - Observing $\boldsymbol{A}\boldsymbol{x} = \sum x_i \boldsymbol{A}_{:,i}$, indicating how far (x_i) we should travel in each of these directions of $\boldsymbol{A}_{:,i}$.
   - In general, this kind of operation is called a linear combination having the formula such that

$$\sum_i c_i \boldsymbol{v}^{(i)}$$

   - The span of a set of vectors is the set of all points obtainable by linear combinations.

Identity and Inverse matrix II

- Linear dependence
  1. Column space of $\boldsymbol{A}$ should be at least m since $\boldsymbol{b}$ can be on the $\mathbb{R}^m$ without any restrictions.
  2. When $n > m$, there can be redundancy, such as the spans of some columns being the same.

$$\exists c_j \neq 0,\ \boldsymbol{A}_{:,i} = \sum_{j \neq i} c_j \boldsymbol{A}_{:,j},$$

  3. Therefore, the matrix $\boldsymbol{A}$ should be square and non-singular for the unique solution. A square matrix with linearly dependent columns is known as singular.
  4. Although $m \neq n$ or singular, solutions can exist.

Norms and Special Kinds of Matrices and Vectors I

1. Norms

- Definition of L^p norm

$$\|\boldsymbol{x}\|_p = \left(\sum_i |x_i|^p \right)^{1/p}$$

- Norm function f satisfies the following:
  1. $f(\boldsymbol{x}) = 0 \Rightarrow \boldsymbol{x} = \boldsymbol{0}$
  2. $f(\boldsymbol{x} + \boldsymbol{y}) \leq f(\boldsymbol{x}) + f(\boldsymbol{y})$ (triangle inequality)
  3. $\forall \alpha \in \mathbb{R}, f(\alpha \boldsymbol{x}) = |\alpha| f(\boldsymbol{x})$
- When $p = 2$, Euclidean norm. Denoted by $\|\boldsymbol{x}\| = \sqrt{\boldsymbol{x}^\top \boldsymbol{x}}$. It is very convenient for the derivative to use the square of the L^2 norm.

Norms and Special Kinds of Matrices and Vectors II

- Note that the L^2 norm provides too large values in machine learning discrimination between zero and non-zero.
- L^1 and L^0 norm are considered to address the above.

$$
\begin{aligned}
\|\boldsymbol{x}\|_1 &= \sum_i |x_i| \\
\|\boldsymbol{x}\|_0 &= \sum_i \mathbb{I}(x_i \neq 0)
\end{aligned}
$$

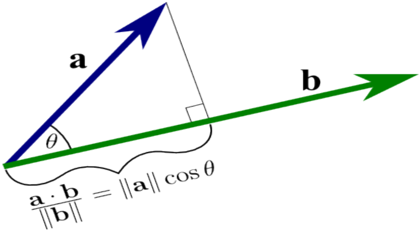
- Other norms such as **max norm** and **Frobenius norm**:

$$
\begin{aligned}
\|\boldsymbol{x}\|_\infty &= \max_i |x_i| \\
\|\boldsymbol{A}\|_F &= \sqrt{\sum_{i,j} A_{i,j}^2}
\end{aligned}
$$

Norms and Special Kinds of Matrices and Vectors III

- Definition of **an angle between two vectors**.

$$\cos\theta = \frac{\boldsymbol{x}^\top \boldsymbol{y}}{\|\boldsymbol{x}\|\|\boldsymbol{y}\|}$$



Norms and Special Kinds of Matrices and Vectors IV

2. Special kinds of matrices and vectors

- **Diagonal matrix**: $D_{i,i} \neq 0, D_{i,j} = 0$ for $i \neq j$.
    1. $\mathrm{diag}(\boldsymbol{v}) \in \mathbb{R}^{n \times n}$ where $\boldsymbol{v} \in \mathbb{R}^n$.
    2. $\mathrm{diag}(\boldsymbol{v})\boldsymbol{x} = \boldsymbol{v} \odot \boldsymbol{x}$.
    3. $\mathrm{diag}(\boldsymbol{v})^{-1} = \mathrm{diag}([1/v_1, \ldots 1/v_n]^\intercal)$.
    4. A diagonal matrix cannot be a square matrix.

Norms and Special Kinds of Matrices and Vectors V

- **Symmetric matrix**: $\boldsymbol{A} = \boldsymbol{A}^\top$.
  1. $A_{i,j} = A_{j,i}$.
  2. Square matrix.
- **Unit vector**: $\|\boldsymbol{x}\| = 1$.
- **Orthogonal matrix**: $\boldsymbol{A}\boldsymbol{A}^\top = \boldsymbol{A}^\top\boldsymbol{A} = \boldsymbol{I}$.

Decompositions I

1. Eigendecomposition

$$\boldsymbol{A} = \boldsymbol{V}\mathrm{diag}(\boldsymbol{\lambda})\boldsymbol{V}^{-1}$$

where $\boldsymbol{V} \in \mathbb{R}^{n\times n}$, $\boldsymbol{\lambda} = [\lambda_1, \ldots, \lambda_n]^\intercal$ is an orthogonal matrix.

- If $\boldsymbol{A}$ is symmetric, then $\boldsymbol{\lambda}$ is a real vector.
- If $\boldsymbol{a}^\intercal \boldsymbol{A}\boldsymbol{a} \geq 0$ for all $\boldsymbol{a} \in \mathbb{R}^n$ (non-negative definite), then $\boldsymbol{\lambda}$ is a non-negative real vector.
- If $\boldsymbol{a}^\intercal \boldsymbol{A}\boldsymbol{a} > 0$ for all $\boldsymbol{a} \in \mathbb{R}^n \setminus \{\boldsymbol{0}\}$ (positive definite), then $\boldsymbol{\lambda}$ is a positive real vector.

Decompositions II

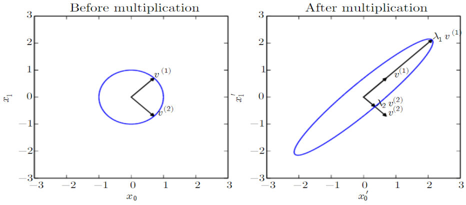


Figure 2.3: An example of the effect of eigenvectors and eigenvalues. Here, we have a matrix $\boldsymbol{A}$ with two orthonormal eigenvectors, $\boldsymbol{v}^{(1)}$ with eigenvalue λ_1 and $\boldsymbol{v}^{(2)}$ with eigenvalue λ_2. *(Left)*We plot the set of all unit vectors $\boldsymbol{u} \in \mathbb{R}^2$ as a unit circle. *(Right)*We plot the set of all points $\boldsymbol{Au}$. By observing the way that $\boldsymbol{A}$ distorts the unit circle, we can see that it scales space in direction $\boldsymbol{v}^{(i)}$ by λ_i.

Figure 1: Deep Learning (Ian Goodfellow and Yoshua Bengio and Aaron Courville, MIT Press book).

Decompositions III

- Some properties
  1. $\max_{\boldsymbol{a},\|\boldsymbol{a}\|=1} \boldsymbol{a}^\intercal \boldsymbol{A}\boldsymbol{a} = \max_i \lambda_i$ where $\lambda_n \leq \lambda_{n-1} \leq \cdots \leq \lambda_1$.
  2. $\operatorname{argmax}_{\boldsymbol{a},\|\boldsymbol{a}\|=1} \boldsymbol{a}^\intercal \boldsymbol{A}\boldsymbol{a} = \boldsymbol{V}_{:,1}$
  3. $\min_{\boldsymbol{a},\|\boldsymbol{a}\|=1} \boldsymbol{a}^\intercal \boldsymbol{A}\boldsymbol{a} = \min_i \lambda_i$.
  4. $\boldsymbol{a}^\intercal \boldsymbol{A}\boldsymbol{a} = \sum_{i=1}^n \lambda_i \boldsymbol{a}^\intercal \boldsymbol{V}_{:,i} \boldsymbol{V}_{:,i}^\intercal \boldsymbol{a} \stackrel{\text{def}}{=} \sum_{i=1}^n \lambda_i \boldsymbol{a}^\intercal \boldsymbol{v}^{(i)} (\boldsymbol{v}^{(i)})^\intercal \boldsymbol{a}$.

Decompositions IV

2. Singular value decomposition

- Decomposition for a non-square matrix $\boldsymbol{A} \in \mathbb{R}^{n \times m}$:

$$\boldsymbol{A} = \boldsymbol{U}\boldsymbol{D}\boldsymbol{V}^{\intercal}$$

where $\boldsymbol{U} \in \mathbb{R}^{n \times n}, \boldsymbol{D} \in \mathbb{R}^{n \times m}$, and $\boldsymbol{V} \in \mathbb{R}^{m \times m}$. Here, $\boldsymbol{D}$ is a diagonal matrix, $\boldsymbol{U}\boldsymbol{U}^{\intercal} = I_n$, and $\boldsymbol{V}\boldsymbol{V}^{\intercal} = I_m$

1. The $\boldsymbol{V}$ matrix of eigendecomposition of $\boldsymbol{A}\boldsymbol{A}^{\intercal}$ is $\boldsymbol{U}$.
2. The $\boldsymbol{V}$ matrix of eigendecomposition of $\boldsymbol{A}^{\intercal}\boldsymbol{A}$ is $\boldsymbol{V}$.

The Moore-Penrose pseudoinverse, trace, and determinant I

1. The Moore-Penrose pseudoinverse

- The Moore-Penrose pseudoinverse is defined as:

$$\boldsymbol{A}^{+} = \lim_{\alpha \downarrow 0} (\boldsymbol{A}^{\intercal}\boldsymbol{A} + \alpha \boldsymbol{I})^{-1}\boldsymbol{A}^{\intercal}.$$

- In fact, $\boldsymbol{A}^{+} = \boldsymbol{U}\boldsymbol{D}^{+}\boldsymbol{V}^{\intercal}$, where $\boldsymbol{D}^{+}$ is obtained by the reciprocal of only non-zero elements of $\boldsymbol{D}^{+}$. Consider $\|\boldsymbol{A}\boldsymbol{x} - \boldsymbol{y}\|^2$.
    1. More columns than rows: $\boldsymbol{x} = \boldsymbol{A}^{+}\boldsymbol{y}$ provides the minimum Euclidean solution among many solutions.
    2. More rows than columns: With $\boldsymbol{x} = \boldsymbol{A}^{+}\boldsymbol{y}$, $\boldsymbol{A}\boldsymbol{x}$ is as close as possible to $\boldsymbol{y}$ in terms of $\|\boldsymbol{A}\boldsymbol{x} - \boldsymbol{y}\|^2$.

The Moore-Penrose pseudoinverse, trace, and determinant II

2. Trace
   - Definition of trace: $\mathrm{Tr}(\boldsymbol{A}) = \sum_i A_{i,i}$.
   - Properties
     1. $\mathrm{Tr}(\boldsymbol{A}) = \mathrm{Tr}(\boldsymbol{A}^\intercal)$.
     2. $\mathrm{Tr}(\boldsymbol{ABC}) = \mathrm{Tr}(\boldsymbol{CAB}) = \mathrm{Tr}(\boldsymbol{BCA})$.
     3. $\|\boldsymbol{A}\|_F = \sqrt{\mathrm{Tr}(\boldsymbol{AA}^\intercal)}$.
3. Determinant
   - $\mathrm{Det}(\boldsymbol{A})$.
   - Product of eigenvalues.
   - Interpreted as a measure of how much multiplication by the matrix expands or contracts space.

Example: PCA I

1. Problem setting
   - Having $\{\boldsymbol{x}^{(1)}, \ldots, \boldsymbol{x}^{(n)}\}$ where $\boldsymbol{x}^{(i)} \in \mathbb{R}^n$.
   - Encoding $f(\boldsymbol{x}) = \boldsymbol{c}$ where $\boldsymbol{c} \in \mathbb{R}^l$ and $l \ll n$.
   - Decoding is $g(f(\boldsymbol{x}))$.
   - Let $g(\boldsymbol{c}) = \boldsymbol{D}\boldsymbol{c}$ where $\boldsymbol{D} \in \mathbb{R}^{n \times l}$.

2. Optimization
   - Restriction: the columns of $\boldsymbol{D}$ are orthogonal, and have unit norms.
   - Optimization:

$$\boldsymbol{c}^* = \text{argmin}_{\boldsymbol{c}} \|\boldsymbol{x} - g(\boldsymbol{c})\|^2.$$

Example: PCA II

Note that $\|\boldsymbol{x} - g(\boldsymbol{c})\|^2 = \boldsymbol{x}^\top \boldsymbol{x} - 2\boldsymbol{x}^\top g(\boldsymbol{c}) + g(\boldsymbol{c})^\top g(\boldsymbol{c})$.

- We have the following:

$$\begin{aligned}
\boldsymbol{c}^* &= \operatorname{argmin}_{\boldsymbol{c}} \|\boldsymbol{x} - g(\boldsymbol{c})\|^2 \\
&= \operatorname{argmin}_{\boldsymbol{c}} g(\boldsymbol{c})^\top g(\boldsymbol{c}) - 2\boldsymbol{x}^\top g(\boldsymbol{c}) \\
&= \operatorname{argmin}_{\boldsymbol{c}} \boldsymbol{c}^\top \boldsymbol{D}^\top \boldsymbol{D}\boldsymbol{c} - 2\boldsymbol{x}^\top \boldsymbol{D}\boldsymbol{c} \\
&= \operatorname{argmin}_{\boldsymbol{c}} \boldsymbol{c}^\top \boldsymbol{c} - 2\boldsymbol{x}^\top \boldsymbol{D}\boldsymbol{c}.
\end{aligned}$$

- Gradient

$$\begin{aligned}
\nabla_{\boldsymbol{c}} \left\{\boldsymbol{c}^\top \boldsymbol{c} - 2\boldsymbol{x}^\top \boldsymbol{D}\boldsymbol{c}\right\} &= 2\boldsymbol{c} - 2\boldsymbol{D}^\top \boldsymbol{x} = \boldsymbol{0} \\
\boldsymbol{c} &= \boldsymbol{D}^\top \boldsymbol{x}.
\end{aligned}$$

Example: PCA III

- Combining the above: $r(\boldsymbol{x}) = g(f(\boldsymbol{x})) = \boldsymbol{D}\boldsymbol{D}^\intercal \boldsymbol{x}$.

$$\boldsymbol{D}^* = \operatorname{argmin}_{\boldsymbol{D}} \sqrt{\sum_{i,j} (x_j^{(i)} - r(\boldsymbol{x}^{(i)})_j)^2} \text{ subject to } \boldsymbol{D}^\intercal \boldsymbol{D} = \boldsymbol{I}_l.$$

- For simplicity, we let $l = 1$.

$$\begin{aligned}
\boldsymbol{d}^* &= \operatorname{argmin}_{\boldsymbol{d}} \sqrt{\sum_i \|\boldsymbol{x}^{(i)} - \boldsymbol{d}\boldsymbol{d}^\intercal \boldsymbol{x}^{(i)}\|^2} \text{ subject to } \boldsymbol{d}^\intercal \boldsymbol{d} = 1 \\
&= \operatorname{argmin}_{\boldsymbol{d}} \sqrt{\sum_i \|\boldsymbol{x}^{(i)} - \boldsymbol{x}^{(i)\intercal} \boldsymbol{d}\boldsymbol{d}\|^2} \text{ subject to } \boldsymbol{d}^\intercal \boldsymbol{d} = 1 \\
&= \operatorname{argmin}_{\boldsymbol{d}} \|\boldsymbol{X} - \boldsymbol{X}\boldsymbol{d}\boldsymbol{d}^\intercal\|_F^2 \text{ subject to } \boldsymbol{d}^\intercal \boldsymbol{d} = 1,
\end{aligned}$$

where $\boldsymbol{X}_{i,:} = \boldsymbol{x}^{(i)\intercal}$.

Example: PCA IV

- Temporarily, we suppress "the subject to".

$$
\begin{aligned}
& \operatorname{argmin}_{\boldsymbol{d}} \|\boldsymbol{X} - \boldsymbol{X}\boldsymbol{d}\boldsymbol{d}^\top\|_F^2 \\
= \quad & \operatorname{argmin}_{\boldsymbol{d}} \operatorname{Tr}\left((\boldsymbol{X} - \boldsymbol{X}\boldsymbol{d}\boldsymbol{d}^\top)^\top(\boldsymbol{X} - \boldsymbol{X}\boldsymbol{d}\boldsymbol{d}^\top)\right) \\
= \quad & \operatorname{argmin}_{\boldsymbol{d}} \operatorname{Tr}\left(\boldsymbol{X}^\top\boldsymbol{X} - \boldsymbol{X}^\top\boldsymbol{X}\boldsymbol{d}\boldsymbol{d}^\top - \boldsymbol{d}\boldsymbol{d}^\top\boldsymbol{X}^\top\boldsymbol{X} + \boldsymbol{d}\boldsymbol{d}^\top\boldsymbol{X}^\top\boldsymbol{X}\boldsymbol{d}\boldsymbol{d}^\top\right) \\
= \quad & \operatorname{argmin}_{\boldsymbol{d}} - 2\operatorname{Tr}\left(\boldsymbol{X}^\top\boldsymbol{X}\boldsymbol{d}\boldsymbol{d}^\top\right) + \operatorname{Tr}\left(\boldsymbol{d}\boldsymbol{d}^\top\boldsymbol{X}^\top\boldsymbol{X}\boldsymbol{d}\boldsymbol{d}^\top\right)
\end{aligned}
$$

- Recap the "subject".

$$
\begin{aligned}
& \operatorname{argmin}_{\boldsymbol{d}} - 2\operatorname{Tr}\left(\boldsymbol{X}^\top\boldsymbol{X}\boldsymbol{d}\boldsymbol{d}^\top\right) + \operatorname{Tr}\left(\boldsymbol{d}\boldsymbol{d}^\top\boldsymbol{X}^\top\boldsymbol{X}\boldsymbol{d}\boldsymbol{d}^\top\right) \text{ subject to } \boldsymbol{d}^\top\boldsymbol{d} = 1 \\
= \quad & \operatorname{argmin}_{\boldsymbol{d}} - 2\operatorname{Tr}\left(\boldsymbol{X}^\top\boldsymbol{X}\boldsymbol{d}\boldsymbol{d}^\top\right) + \operatorname{Tr}\left(\boldsymbol{X}^\top\boldsymbol{X}\boldsymbol{d}\boldsymbol{d}^\top\right) \text{ subject to } \boldsymbol{d}^\top\boldsymbol{d} = 1 \\
= \quad & \operatorname{argmax}_{\boldsymbol{d}} \operatorname{Tr}\left(\boldsymbol{X}^\top\boldsymbol{X}\boldsymbol{d}\boldsymbol{d}^\top\right) \text{ subject to } \boldsymbol{d}^\top\boldsymbol{d} = 1 \\
= \quad & \operatorname{argmax}_{\boldsymbol{d}} \operatorname{Tr}\left(\boldsymbol{d}^\top\boldsymbol{X}^\top\boldsymbol{X}\boldsymbol{d}\right) \text{ subject to } \boldsymbol{d}^\top\boldsymbol{d} = 1
\end{aligned}
$$

- The first PC is the first eigenvector correspondent to the first eigenvalue.